Was magnitude estimation necessary?

A secondary-data reassessment of Bard, Robertson, and Sorace (1996)

Brett Reynolds *
Humber Polytechnic & University of Toronto

June 2026

Abstract

TODO: Draft after the source and dataset inventory is verified. The likely contribution is a no-new-data reassessment of the Bard, Robertson, and Sorace measurement program: acceptability judgments are graded, measurable, and methodologically non-trivial, but the case for magnitude estimation as a method-specific solution to scale problems needs to be separated from the stronger reception claim that it has privileged measurement status. The paper should distinguish reproduction, secondary analysis, conceptual replication, and theoretical revision.

Keywords: magnitude estimation; acceptability judgments; grammaticality judgments; secondary data; experimental syntax

1 The problem

TODO: Open from the narrow methodological question. Bard, Robertson, and Sorace made acceptability judgments experimentally tractable by treating them as graded data, but the lasting contribution may be the formalization of judgment methodology rather than magnitude estimation itself. The paper has to ask what later data shows about that distinction.

2 The Bard target

TODO: Reconstruct the target claims before criticizing them. The candidate claims are scale resolution, fine-grained distinction, use with linguistically naive participants, replication across groups, and a stronger measurement status than simpler rating tasks. Do not depend on original participant-level data unless it's found.

3 Secondary-data design

The secondary-data design treats the project as a measurement reassessment of Bard, Robertson, and Sorace (Bard et al., 1996). Four evidential routes should stay separate. Published summaries from Bard et al. can source the historical claim, but not a direct replication unless their participant-level data are found.

*Contact: brett.reynolds@humber.ca

Later judgment studies can support secondary analysis when they compare elicitation methods on shared or comparable items and provide the rows needed for the intended estimand. Computational benchmarks can show the afterlife of acceptability labels, but they do not bear on magnitude estimation versus Likert ratings. Construct-level discussion can separate acceptability from grammaticality, naturalness, standardness, and correction.

This creates a genre gate. Public row-level data from the Sprouse comparison studies can support a constrained secondary analysis of method performance. The Langsford et al. reliability study is essential evidence for the argument, but unless its raw participant responses are obtained it can only constrain the paper at the article-summary level. The paper should therefore not promise a full response-style or test-retest model across the whole literature.

The Sprouse files also require representation choices before modelling. For Sprouse et al. (2013), the primary forced-choice comparison should use the pair-level sign-test file, because forced choice is a pairwise task and that file already records judgments against expected patterns. The logistic file preserves item-level binary rows, changing the unit of analysis, so it belongs as a named sensitivity specification.

For Sprouse and Almeida (2017), yes/no rows with item IDs `B`, `G`, and `M` should be excluded from item-level comparisons. They are absent from the materials workbook, occur in repeated early order positions for every subject, and do not align with the magnitude-estimation, Likert, or forced-choice item universe. This exclusion is a structural inclusion rule, not an outcome-based trimming decision.

Before any modelling, each source should be assigned to an estimand it can actually answer. The primary estimand is the item-level acceptability signal recoverable across methods. Secondary estimands are method-specific noise or bias and decision agreement for theoretically relevant contrasts. Participant response style and method-by-item instability are admissible only when the dataset contains the participant-level rows and repeated structure needed to estimate them. This ordering keeps the analysis from selecting whichever scale looks best after the fact, the kind of forking path Gelman and Loken warn about (Gelman & Loken, 2013, 2014).

The substantive comparison is a common-latent-signal null: magnitude estimation, Likert ratings, forced choice, and Thurstonian choice are first modelled as noisy observations of the same item-level acceptability. A magnitude-estimation term becomes interesting only if it improves prediction, reliability, or decision stability enough to matter. Scrambled labels, method-label permutations, and simulations from a no-method-advantage latent model are negative controls for the pipeline, checking whether the analysis manufactures a method winner from degraded structure. They are not substitutes for the substantive common-signal comparison.

4 Scale and reliability

The scale question should be phrased as practical measurement advantage over the common-latent null. Bard et al.'s case for magnitude estimation included scale resolution, fine-grained distinctions, naive-participant use, and replication across groups (Bard et al., 1996). The safer target is that source-grounded claim, together with the stronger reading of magnitude estimation that took hold in parts of the later methodology literature. Later method-comparison data give the paper a sharper question: does magnitude estimation recover item differences, participant agreement, or theoretically relevant decisions that simpler tasks lose?

Sprouse et al. (2013) compare formal judgment tasks with informal published judgments, giving evidence about convergence between magnitude estimation, Likert ratings, and forced choice. The public Sprouse rows can support a bounded reanalysis of item-level agreement and decision consequences, subject to reuse permissions. Langsford et al. (2018) add a reliability and response-bias target by comparing Likert, magnitude estimation, forced choice, and Thurstonian models. Without their raw rows, Langsford can support the synthesis of published reliability findings but not a new participant-level response-style or test-retest model.

The reliability section should inherit the analysis charter rather than reopen method choices. In the 2013 data, pair-level forced choice is primary and the item-level logistic representation is sensitivity. In the 2017 data, yes/no rows with non-material IDs `B`, `G`, and `M` are outside the item-level comparison set. These choices make later reliability claims traceable to fixed units of analysis instead of to whichever representation later proves most convenient.

The primary reliability analysis should compare methods on prediction, stability, decision consequences, mean separation, and correlation, with priority on consequences for the linguistic contrast. A useful result would show whether magnitude estimation changes the answer for the linguistic contrast under study, reduces Type S or Type M risks (Gelman & Carlin, 2014), or carries costs that offset any added resolution. Sensitivity checks should vary transformations and exclusion rules as named specifications rather than as invisible post hoc choices (Steegen et al., 2016).

5 The contemporary afterlife of acceptability

TODO: Treat modern benchmark resources as an afterlife of acceptability, not as direct magnitude-estimation evidence. Resources such as BLiMP (Warstadt et al., 2020) and the Syntactic Acceptability Dataset (Juzek, 2024) show how acceptability became portable for computational evaluation, but they also change the object being measured.

6 Grammaticality and acceptability

TODO: Connect the reassessment to the broader *(Un)grammatical* program. The paper should separate personal production, acceptability, naturalness, standardness, correction behaviour, and theoretical grammaticality. Existing datasets may show the boundaries of what no-new-data work can settle.

7 Conclusion

TODO: Conclude with the balanced update. Bard et al. were right to make acceptability measurable and methodologically serious. The case for magnitude estimation as a solution to scale-resolution problems is a separate, source-grounded claim. The later, stronger reading of magnitude estimation as the privileged measurement tool should be reassessed as reception unless Bard is cited directly for it.

Data and code availability

No data has been committed or analysed in this setup draft. The intended project uses reusable public datasets where licensing and documentation permit it; source and dataset verification are tracked in `notes/source-verification.md`.

Acknowledgements

TODO: Add final AI-use disclosure with the actual model list used on the project.

References

Bard, E. G., Robertson, D., & Sorace, A. (1996). Magnitude estimation of linguistic acceptability. *Language*, *72*(1), 32–68. https://doi.org/10.2307/416793

Gelman, A., & Carlin, J. (2014). Beyond power calculations: Assessing type S (sign) and type M (magnitude) errors. *Perspectives on Psychological Science*, *9*(6), 641–651. https://doi.org/10.1177/1745691614551642

Gelman, A., & Loken, E. (2013). *The garden of forking paths: Why multiple comparisons can be a problem, even when there is no "fishing expedition" or "p-hacking" and the research hypothesis was posited ahead of time* [Unpublished manuscript, Columbia University]. https://sites.stat.columbia.edu/gelman/research/unpublished/p_hacking.pdf

Gelman, A., & Loken, E. (2014). The statistical crisis in science. *American Scientist*, *102*(6), 460–465. https://doi.org/10.1511/2014.111.460

Juzek, T. S. (2024). The Syntactic Acceptability Dataset (preview): A resource for machine learning and linguistic analysis of English. *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, 16113–16120.

Langsford, S., Perfors, A., Hendrickson, A. T., Kennedy, L. A., & Navarro, D. J. (2018). Quantifying sentence acceptability measures: Reliability, bias, and variability. *Glossa: a journal of general linguistics*, *3*(1), Article 37, 1–34. https://doi.org/10.5334/gjgl.396

Sprouse, J., & Almeida, D. (2017). Design sensitivity and statistical power in acceptability judgment experiments. *Glossa: A Journal of General Linguistics*, *2*(1), 1–32. https://doi.org/10.5334/gjgl.236

Sprouse, J., Schütze, C. T., & Almeida, D. (2013). A comparison of informal and formal acceptability judgments using a random sample from *Linguistic Inquiry* 2001–2010. *Lingua*, *134*, 219–248. https://doi.org/10.1016/j.lingua.2013.07.002

Steegen, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing transparency through a multiverse analysis. *Perspectives on Psychological Science*, *11*(5), 702–712. https://doi.org/10.1177/1745691616658637

Warstadt, A., Parrish, A., Liu, H., Mohananey, A., Peng, W., Wang, S., & Bowman, S. R. (2020). BLiMP: The benchmark of linguistic minimal pairs for English. *Transactions of the Association for Computational Linguistics*, *8*, 377–392. https://doi.org/10.1162/tacl_a_00321